

* Testing for Non-stationary - Dickey-Fuller

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t$$

↑
Drift

$H_0: \rho = 1$
 $H_1: \rho < 1$
 Non Stationary Stationary

$$X_t - X_{t-1} = \alpha + (\rho - 1) X_{t-1} + \varepsilon_t$$

$$\Delta X_t = \alpha + \delta X_{t-1} + \varepsilon_t$$

↑

t statistic $\hat{\delta}$, $t < DF_{critical}$

$\Rightarrow$ reject null hypothesis.

* Augmented DF Test

AR(1)

$$\Delta y_t = \alpha + \delta y_{t-1} + \varepsilon_t$$

$t < DF_{critical}$
 $-ve$

No serial correlation
↓

$$\Rightarrow \Delta y_t = \alpha + \delta y_{t-1} + \sum_{i=1}^n \beta_i \Delta y_{t-i} + \varepsilon_t$$

$H_0: \delta = 0$, non-stationary

$H_1: \delta < 0$, station

↑ t dist.
F tests

* Kalman Filter

1) Introduction

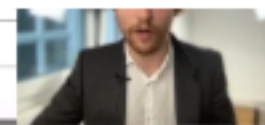
Mathematical method for iteratively estimating parameters to describe system states based on error prone observation.

2) Intuition behind Kalman Filter

- ⇒ It predicts first, sees actual ans., fixes it and then use it to predict next.
- ⇒ It is recursive - new measurements can be processed as they are measured.
- ⇒ Optimal estimator in case of all noise is Gaussian.
- ⇒ Minimizes the error function (mean sq. error) of the estimated parameters.

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HUDSON & THAMES | TIME SERIES ANALYSIS: KALMAN FILTERS



Visual example

Estimating a car's actual position

- Assuming the noise of the position measurement of the car is gaussian distributed.
- Use filter to give an optimal state estimate.
- Given our initial state estimate, we predict the next time step.
- After a measurement at time step k the prediction is updated.

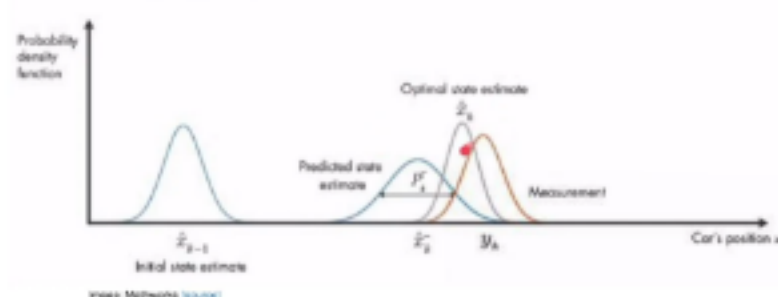
Prediction

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$
$$P_k^- = AP_{k-1}A^T + Q$$

Initial estimates for $\hat{x}_{k-1}$ and P_{k-1}

Update

$$K_k = \frac{P_k^- C^T}{CP_k^- C^T + R}$$
$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - C\hat{x}_k^-)$$
$$P_k = (I - K_k C)P_k^-$$





Prediction step

Extrapolation, prediction of system values

1.1 The filter algorithm consists of a prediction step

$$\mathbf{x}_{k|k-1} = \mathbf{F}_{k-1} \mathbf{x}_{k-1} + \mathbf{B}_{k-1} \mathbf{u}_{k-1}$$

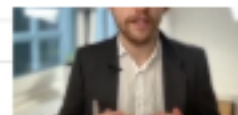
1.2 Then the covariance error of the system is predicted

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1} \mathbf{P}_{k-1} \mathbf{F}_{k-1}^T + \mathbf{Q}_{k-1}$$

$\mathbf{x}_{k k-1}^{(n)}$	Predicted system state at time step k , before applying the new measurement.
$\mathbf{F}_{k-1}^{(n \times n)}$	Transition matrix that propagates the system state to next time step.
$\mathbf{B}_{k-1}^{(n \times d)}$	Deterministic disturbance.
$\mathbf{u}_{k-1}^{(d)}$	Dynamics of the deterministic disturbance and projection onto the system state.
$\mathbf{P}_{k k-1}^{(n \times n)}$	Covariance matrix of the errors of $\mathbf{x}$ at time step k , before applying the new observations
$\mathbf{Q}_{k-1}^{(n \times n)}$	Process noise, describes additional uncertainties due to modelling errors.



8



Update step

Update the values of the system

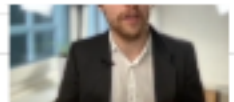
2.1. First, the actual system value is measured with z at time step k and then the innovation or I is calculated

$$\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{H}_k \mathbf{x}_{k|k-1}$$

2.2 Then, the covariance error of the measurement is calculated

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k$$

$\mathbf{H}_k^{(m \times n)}$	Observation matrix
$\mathbf{R}_k^{(m \times m)}$	Covariance matrix of the measurement noise



Update step

Update the values of the system

2.3. First, the Kalman gain is calculated at time step k

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1}$$

2.4. Now, we can update the system value (give an estimate)

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \tilde{\mathbf{y}}_k$$

2.4. Lastly, the covariance error at time step k is calculated / estimated

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1}$$

Prediction	→	Update
$\mathbf{x}_{k k-1} = \mathbf{F}_{k-1} \mathbf{x}_{k-1} + \mathbf{B}_{k-1} \mathbf{u}_{k-1}$		$\mathbf{K}_k = \mathbf{P}_{k k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k k-1} \mathbf{H}_k^T + \mathbf{R}_k)^{-1}$
$\mathbf{P}_{k k-1} = \mathbf{F}_{k-1} \mathbf{P}_{k-1} \mathbf{F}_{k-1}^T + \mathbf{Q}_{k-1}$		$\mathbf{x}_k = \mathbf{x}_{k k-1} + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H}_k \mathbf{x}_{k k-1})$
		$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k k-1}$

$\mathbf{K}_k^{(m \times n)}$	Kalman gain matrix for projecting the residuals onto the correction of the system state.
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3) Using in Finance

⇒ Assuming returns are following a gaussian distribution, it can be theoretically used to smooth out the data.

⇒ Estimates the mean of the price distribution at time step k which is our optimal estimate.

⇒ Compared to moving avg there is no need to specify day lag. Recent obs. are more heavily weighted.

4) Limits

- 1) When treating every small movement as meaningful, we won't filter out the noise.
- 2) If we smooth too much, might lose too much info.

Shumway Stoffer Smoother

Published in 1982 by Robert H. Shumway, David S. Stoffer

Abstract: An approach to smoothing and forecasting for time series with missing observations is proposed. For an underlying state-space model, the EM algorithm is used in conjunction with the conventional Kalman smoothed estimators to derive a simple recursive procedure for estimating the parameters by maximum likelihood. An example is given which involves smoothing and forecasting an economic series using the maximum likelihood estimators for the parameters.

5) Likelihood Function

Given a prob. density function or count density

$$f: \mathbb{R}^n \longrightarrow \mathbb{R}$$

which is additionally influenced by one or more param. ϑ from a set Θ .

$$L: \Theta \longrightarrow \mathbb{R}$$

This gives us the Likelihood function.

$$L(\vartheta) = f_{\vartheta}(x)$$

=> For example,
prob. of something in gaussian distribution.

=> Max^m Likelihood Function

=> A method estimating the param. of prob. distribution

$$f: \Omega \rightarrow [0, 1], \quad x \mapsto f(x|\vartheta)$$

$$L: \Theta \rightarrow [0, 1], \quad \vartheta \mapsto f(x|\vartheta)$$

Shumway Stoffer Smoother

Intuition

- Given we know the parameters θ of our model, we calculate the likelihood function.
- Given we know the likelihood function, we can estimate the parameters with the maximum likelihood method.
- What if need to guess our parameters?

EM Smoothing Algorithm

Input $y = (y_0, y_1, \dots, y_N)$
 Initialise $\theta_0 = (A_0, B_0, C_0, D_0)$, $\hat{x}_0 = y_0$ and $\Sigma_0 = D^2$
for $j = 0, 1, \dots$ **do**
 Apply Kalman Filter with Σ_0 , $\hat{x}_0$ and θ_j
 E-Step: Apply Kalman Smoothing with $\hat{x}_{N|N}$ and $\Sigma_{N|N}$ to compute $Q(\theta, \theta_j)$
 M-Step: Compute parameter updates as $\theta_{j+1} = \arg \max Q(\theta, \theta_j)$
 Update initial filter values for the following iteration as $\hat{x}_0 = \hat{x}_{0|N}$ and $\Sigma_0 = \Sigma_{0|N}$
 Continue until $\mathcal{L}(y|\theta_{j+1}) - \mathcal{L}(y|\theta_j) \leq \text{tolerance}$ or $j = \text{max iterations}$
Return θ

Shumway Stoffer Smoother

Implementation in the ArbitrageLab

- Multi-functional with a wide range of application
- Provides predictions when working with missing data
- Provides covariance interval
- Results tested by code of the author of the paper

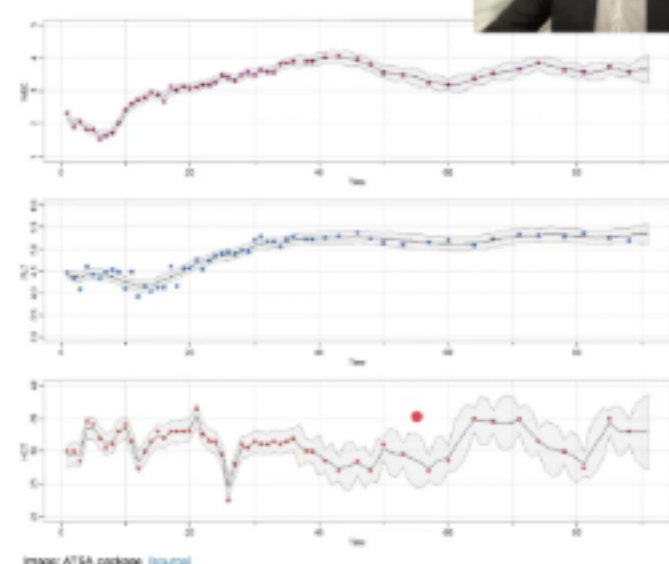


Image: ATSA package [\[source\]](#)

6) Pairs Trading

⇒ Basically, if we have two random walks, they might be correlated and form a mean reverting.

⇒ If we long on one and short on other, we would be trading on the diff. b/w these two.

7) Applying Kalman Filter for trading on spread.

Applying the Kalman filter for trading the spread

Using the Shumway Stoffer Smoother and generate signals for trading

Given the process is mean reverting, we can use the methods described in the paper by Elliot et al (2015) to generate trading signals given the outputs of the Kalman filter:

$$u_k = \begin{cases} \text{short } r_k \text{ units of high and long 1 unit of low (long spread trade)} & \text{if } \frac{x_k - \hat{x}_{k|k-1}}{\hat{\sigma}_{k|k-1}} \geq \alpha \\ \text{short } r_k \text{ units of low and long 1 unit of high (short spread trade)} & \text{if } \frac{x_k - \hat{x}_{k|k-1}}{\hat{\sigma}_{k|k-1}} \leq -\alpha \\ \text{no trade} & \text{otherwise} \end{cases}$$

Conclusion

Exploring the Kalman filter

- Versatile method for fast signal processing
- We can estimate parameters using the EM-algorithm
- Performance with real data varies
- Limitation can be that the noise is not gaussian distributed
- Trading signals need a parameter (alpha)

